

Physics 7: Midterm 1

Name: _____

Student ID: _____

Show your work. Answers without work get partial credit at most.

1. (12 points) A ball is dropped from 20 m. How long does it take to land?

2. (8 points) State Newton's second law in words.

3. A block slides down a frictionless ramp.

(a) (4 points) Draw the forces on the block.

(b) (6 points) Find its acceleration on a 30 degree ramp.

4. (15 points) Explain why the sky is blue.

5. (5 points) Name one quantity conserved in an elastic collision.